

Tutorial 9 Soft Margin SVM (9.4)

primal problem: $\min_{\{w, b, \xi\}} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$

inverse of margin
slack variable
penalty

s.t. $y_i (w^T x_i + b) \geq 1 - \xi_i, \forall_i$
margin constraint
 $\xi_i \geq 0$, violate the margin
slack variable $\xi_i \geq 0$

a) let $d_i \geq 0$ be lagrange multiplier for $\xi_i \geq 0$

lagrange func.

$$L(w, b, \xi, d, r) = \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

for $\lambda g(x)$ objective

$$- \sum_i d_i (y_i (w^T x_i + b) - 1 + \xi_i)$$

$$- \sum_i r_i \xi_i$$

b) at the min of L , derivative w.r.t $\{w, b, \xi\}$, set as 0

$$\frac{\partial}{\partial w} L = w - \sum_i d_i y_i x_i = 0 \Rightarrow w^* = \sum_i d_i y_i x_i$$

$$\frac{\partial}{\partial b} L = - \sum_i d_i y_i = 0 \Rightarrow \sum_i d_i y_i = 0$$

$$\frac{\partial}{\partial \xi_i} L = C - d_i - r_i = 0 \Rightarrow r_i + d_i = C$$

$$c) L(w, r) = \min_{w, b, \xi} L(w, b, \xi, d, r)$$

$$= \frac{1}{2} \left\| \sum_i d_i y_i x_i \right\|^2 + C \sum_i \xi_i - \sum_i d_i (y_i (\sum_j d_j y_j x_j^T x_i + b) - 1 + \xi_i)$$

$$= \frac{1}{2} \left(\sum_i d_i y_i x_i \right)^T \left(\sum_j d_j y_j x_j \right) - \sum_i \sum_j d_i d_j y_i y_j x_i^T x_j$$

$$+ \sum_i (C - d_i - r_i) \xi_i - \sum_i d_i y_i b + \sum_i d_i$$

$$= \sum_i d_i - \frac{1}{2} \sum_i \sum_j d_i d_j y_i y_j x_i^T x_j = L(d)$$

d) constraints: $\begin{cases} d_i \geq 0 \\ r_i \geq 0 \\ d_i = C - r_i \end{cases} \Rightarrow 0 \leq d_i \leq C$

dual SVM

$$\max_d \sum_i d_i - \frac{1}{2} \sum_i \sum_j d_i d_j y_i y_j x_i^T x_j$$

s.t. $0 \leq d_i \leq C \quad \forall_i$

$$\sum_i d_i y_i = 0$$

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e) KKT conditions

LM

$$y_i (W^T x_i + b) > 1 - \xi_i$$

α_i

$$\xi_i \geq 0$$

γ_i

$$W^* = \sum_i \alpha_i y_i x_i$$

on the margin $0 < \alpha_i < C$

violate margin $\alpha_i = C$

1) $\alpha_i = 0, \xi_i = 0 \Rightarrow y_i (W^T x_i + b) > 1 \Rightarrow$ correctly classified outside margin

$$\xi_i > 0$$

2) $0 < \alpha_i < C, \xi_i = 0 \Rightarrow y_i (W^T x_i + b) = 1 \Rightarrow$ correct ✓
on the margin.

3) $\alpha_i = C \Rightarrow \xi_i = 0 \Rightarrow \xi_i > 0 \Rightarrow y_i (W^T x_i + b) = 1 - \xi_i$
inside (violate) margin
if $\xi_i > 1, \Rightarrow$ misclassified

